# O-LEVEL CHEMISTRY

## INTRODUCTION TO CHEMISTRY

Chemistry is a branch of science that deals with the study of structure, composition and properties of matter. Chemistry as an experimental subject relies on:
- Careful handling of apparatus
- Carrying out chemical tests or chemical analysis
- Making critical observation
- Critical reporting of observations
- Drawing appropriate conclusions from observations

Chemistry as a practical science tries to answer the following question.
What are materials made up of?
How is matter formed?
How does matter behave?
Why does matter behave the way it does?

There are quite a number of branches of chemistry such as Organic chemistry, Physical chemistry, Inorganic Chemistry and Analytical chemistry.

### Importance of chemistry
- The knowledge of chemistry makes us to understand the properties of substances and handle these substances with great care. Some may be poisonous, corrosive, toxic etc.
- Makes us to understand how to extract substances from the earth and use them e.g. gold, copper etc.
- Chemistry knowledge opens up way to science professional courses such as medicine, education and pharmacy.
- The knowledge of chemistry makes us to know the effects of chemicals on the environment and subsequently puts us in position to protect our environment.
- Broad knowledge in chemistry puts us in better positions to contribute towards more advancement in science and technology for better and quality human life.
- The knowledge of chemistry is relevant in many ways such as making of food supplements, distillation of fuel, making of plastics, making cosmetics and dental creams, manufacture of soap and detergents, making insecticides and herbicides.etc.

### LABORATORY
A laboratory is a special place with special equipments where scientific investigations or experiments are carried out. The laboratory is a place of adventure and discovery; in fact some of the most exciting events in the history of science have taken place in laboratories e.g. the discovery of oxygen.
When experiments are carried out in the laboratory, chemical materials and are used.

**Chemicals** are substances usually in solid or liquid forms that are consumed during reactions. They get usedup in reactions during the course of an experiment. E.g. Sodium hydroxide and Hydrochloric acid.